

Week 3 Task Update Report (Final)

Progress Completed:

- Successfully wrote the code for the **Calibration System**. When the node starts, it now takes 2 seconds to read the phone's resting angle and sets that exact position as "zero."
- Built a complete **Telemetry Visualization Dashboard** using HTML and JavaScript.
- Connected the web browser to the ROS 2 environment using `rosbridge_suite`.
- The web dashboard successfully displays live, animated bars showing the robot's forward and turning speeds in real-time!
- Recorded the final demonstration video and drew the system architecture diagram.
- **All minimum requirements and all bonus objectives have been 100% completed.**

Challenges Faced:

- **Connecting ROS to a Web Browser:** Getting a standard website to talk to a robotics system was tricky because they use different network rules. I solved this by learning how to use the `rosbridge_server` package, which translates ROS 2 data into WebSockets that my HTML dashboard can easily read.

Plans for the Next Week:

- Since this is the final week, I have no technical plans for next week! My next step is to submit the final Google Form with my GitHub repository link, the demonstration video, and my documentation.